🤌 XAI Lecture 13

Explainable Medical Coding & Attention is not Explanation

FAMAF - UNC

First Semester 2026

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

🧉 Paper 1 ---

Explainable Prediction of Medical Codes from Clinical Text

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Abstract

Clinical notes are text documents that are created by clinicians for each patient encounter. They are typically accompanied by medical codes, which describe the diagnosis and treatment. Annotating these codes is labor intensive and error prone; furthermore, the connection between the codes and the text is not annotated, obscuring the reasons and details behind specific diagnoses and treatments. We present

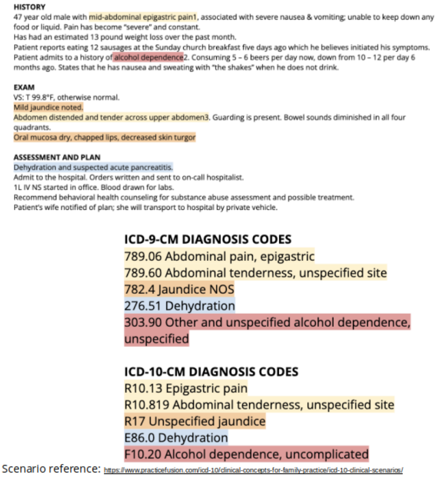
and over 140,000 codes combined in the newer ICD-10-CM and ICD-10-PCS taxonomies (World Health Organization, 2016). Second, clinical text includes irrelevant information, misspellings and non-standard abbreviations, and a large medical vocabulary. These features combine to make the prediction of ICD codes from clinical notes an especially difficult task, for computers and human coders alike (Birman-Deych et al., 2005).

In this application paper, we develop convolu-

Figure 1: alt text

🧉 Introduction ---

- **Clinical notes**: free text narratives generated by clinicians during patient encounters
- **ICD codes**: metadata codes accompanying clinical notes
  - Standardized way of indicating diagnoses and procedures
  - Multiple use cases (billing, predictive modelling)



HISTORY

47 year old male with mid-abdominal epigastric pain1, associated with severe nausea & vomiting; unable to keep down any food or liquid. Pain has become "severe" and constant.
Has had an estimated 13 pound weight loss over the past month.
Patient reports eating 12 sausages at the Sunday church breakfast five days ago which he believes initiated his symptoms.
Patient admits to a history of alcohol dependence2. Consuming 5 – 6 beers per day now, down from 10 – 12 per day 6 months ago. States that he has nausea and sweating with "the shakes" when he does not drink.

EXAM

VS: T 99.8°F, otherwise normal.
Mild jaundice noted.
Abdomen distended and tender across upper abdomen3. Guarding is present. Bowel sounds diminished in all four quadrants.
Oral mucosa dry, chapped lips, decreased skin turgor

ASSESSMENT AND PLAN

Dehydration and suspected acute pancreatitis.
Admit to the hospital. Orders written and sent to on-call hospitalist.
1L IV NS started in office. Blood drawn for labs.
Recommend behavioral health counseling for substance abuse assessment and possible treatment.
Patient's wife notified of plan; she will transport to hospital by private vehicle.

ICD-9-CM DIAGNOSIS CODES

789.06 Abdominal pain, epigastric
789.60 Abdominal tenderness, unspecified site
782.4 Jaundice NOS
276.51 Dehydration
303.90 Other and unspecified alcohol dependence, unspecified

ICD-10-CM DIAGNOSIS CODES

R10.13 Epigastric pain
R10.819 Abdominal tenderness, unspecified site
R17 Unspecified jaundice
E86.0 Dehydration
F10.20 Alcohol dependence, uncomplicated

Scenario reference: https://www.practicefusion.com/icd-10/clinical-concepts-for-family-practice/icd-10-clinical-scenarios/

In Plain English

Given a doctor's note, can a model automatically assign the right diagnostic codes — and point to the exact sentence in the text that justifies each one?

Motivation ---

- **Limitation of ICD code annotation**
  - Labor intensive
  - Coders read 720–1,440 pages of clinical documentation per day
  - Prone to error
  - Lack of connection bridging text and code
- **Challenges of automatic coding**
  - Features of clinical text: irrelevant information, misspellings, non-standard abbreviations, large medical vocabulary
  - Label space

🧉 Summary of Contributions ---

- **CAML**: Convolutional Attention for Multi-Label classification
  - Attentional convolutional network: CNN + per-label attention mechanism
  - Important information may be in short snippets anywhere in the document
  - Decision support in clinical domain
- **Accuracy Evaluation**
  - Used open-access MIMIC datasets
  - Outperformed previously proposed approaches
- **Interpretability Evaluation**
  - Physician rated the informativeness of automatically generated explanations

🧉 Related Work: Automatic ICD Coding ---

- Structured text v. unstructured text
- **Datasets**
  - Subset of full ICD label space (Wang et al., 2016)
  - Subset of medical scenarios (Zhang et al., 2017)
  - Death certificates in English + French (Névéol et al., 2017)
  - Private datasets (Subotin and Davis, 2016)
- **Approaches**
  - Recurrent networks with HA-GRU (Baumel et al., 2018)
  - Character-aware LSTMs (Shi et al., 2017)
  - Memory networks for top-50/100 codes (Prakash et al., 2017)
  - Grounded RNN for individual labels (Vani et al., 2017)

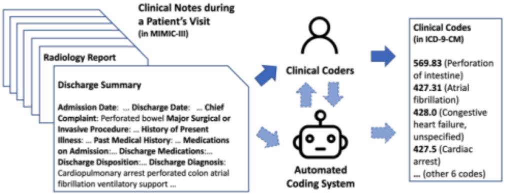


Image reference: https://www.nature.com/articles/s41746-022-00705-7.pdf

CAML instead operates on the full ICD-9 label space using MIMIC-III, an open-access dataset.

🧉 Related Work: Explainable Text Classification ---

- **Latent rationales** (Lei et al., 2016): learn a latent variable that tags each word as relevant or irrelevant to the predicted label
- **Gradient-based salience** (Li et al., 2016): measure word importance by how much each word affects the output
- **Attention as explanation** (Rush et al., 2015; Rocktäschel et al., 2016): highlight salient text features using attention weights

Key Gap

Prior work identifies *which* words the model focuses on, but never verifies whether **attending to completely different words** would have changed the prediction — leaving the causal link between attention and output unexamined.

🧉 Related Work: Attentional Convolution for NLP ---

- CNNs applied to sentiment classification (Kim, 2014) and language modeling
- Convolution and attention: Allamanis et al., 2016; Yin et al., 2016; dos Santos et al., 2016; Yin and Schütze, 2017
- Most relevant work (Yang et al., 2016 and Allamanis et al., 2016):
  - Use context vectors (one for document) to compute attention over specific locations in the text

CAML Differentiation

- Goal: select locations most important for predicting specific labels
- Compute separate attention weights for each label in the label space

🧉 ICD Code Prediction Problem ---

Clinical Text

A 12-year old girl with known hyperagglutinability, presented to the emergency department with a 2-week history of headeaches and facial weakness. Neurologic examination indicated sensorineural hearing loss on the right side with Weber's test lateralizing to the left, and the Rinne's test demonstrating bone conduction greater than air conduction on the right. Magnetic resonance imaging of the head revealed severe structural defects of the right petrous temporal bone. No indication of cerebral infarction.

ICD Codes

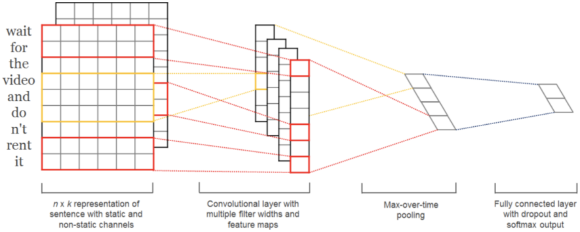
Diagnosis	ICD-9	ICD-10
Cervical Sprain, initial encounter	847.0	**S13.4xxA**
Thoracic Sprain, initial encounter	847.1	**S23.3xxA**
Lumbar Sprain, initial encounter	847.2	**S33.5xxA**
Cervical Degenerative Disc Disease	722.4	**M50**
Thoracic Degenerative Disc Disease	722.51	**M51**
Lumbar Degenerative Disc Disease	722.52	**M51.2**

Multi-Label Classification Problem

- "Match" snippets of clinical text to a label — each code is triggered by a **different part** of the document, not the document as a whole
- For rare codes with few training examples, use the **code's own text description** as an extra learning signal (e.g., "Cervical Degenerative Disc Disease" tells the model what kind of text should activate it)


"IT'S
NEVER
LUPUS."
peacock
HOUSE
M.D.

🧉 Typical Convolutional N-Gram Model ---

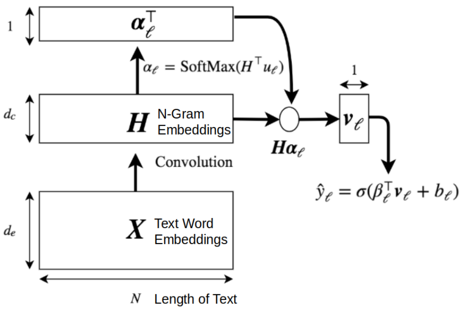


- Each word is represented as a dense vector (**word embedding**)
- A convolutional filter slides over **N consecutive words** at a time, detecting local patterns (e.g., "acute renal failure")
- Each filter learns to recognize a specific **"word snippet" motif**
- **Max pooling** asks: *was this pattern detected anywhere in the document?* — collapsing position information in the process

Key Limitation

Max pooling discards *where* the pattern appeared — you know a pattern was detected, but you can't point to the sentence that triggered it, making the prediction unexplainable to a physician.

Convolutional Attention Model ---



Key Insight

Because $\mathbf{u}_\ell$ is different for every label, each code gets its own attention distribution — and therefore its own extractable text snippet as explanation.

1. **Convolution: $\mathbf{X} \to \mathbf{H}$**
   Each column of $\mathbf{H}$ is an n-gram embedding — a single vector capturing the meaning of N consecutive words in context.
2. **Per-label attention: $\alpha_\ell = \text{SoftMax}(\mathbf{H}^\top \mathbf{u}_\ell)$**
   For each label ℓ, compare every n-gram against the label vector $\mathbf{u}_\ell$ via dot product, then normalize.
   Result: a distribution over document positions — *which fragment is most relevant for this code?*
3. **Weighted aggregation: $\mathbf{v}_\ell = \sum_{n=1}^{N} \alpha_{\ell,n} \mathbf{h}_n$**
   Instead of max pooling, sum all n-grams weighted by attention — preserving *where* the evidence came from.
4. **Classification: $\hat{y}_\ell = \sigma(\beta_\ell^\top \mathbf{v}_\ell + b_\ell)$**
   Linear layer + sigmoid gives the probability for each code independently.

🧉 Training Loss - Binary Cross-Entropy (CAML) ---

$$L_{\mathbf{BCE}}(\mathbf{X}, y) = -\sum_{\ell=1}^{\mathcal{L}} [y_\ell \log(\hat{y}_\ell) + (1 - y_\ell) \log(1 - \hat{y}_\ell)]$$

In Plain English

Each code is treated as an **independent binary classification** — present or not — and errors are accumulated over all ~9,000 possible codes.

🧉 Training Loss Function - With Description Regularization (DR-CAML) 2/2 ---

$$L(\mathbf{X}, y) = L_{\mathbf{BCE}} + \lambda \frac{1}{n_y} \sum_{\ell:\, y_\ell = 1} \|\mathbf{z}_\ell - \beta_\ell\|_2$$

- n_y: number of true labels for data example
- $\mathbf{z}_\ell$: representation of label based on code description
- β_ℓ: representation of label learned by model

In Plain English

For common codes, learn from data. For rare codes, stay close to what the medical dictionary says the code means.

🧉 Experimental Setup ---

Dataset: MIMIC-III

- **Input:** discharge summary (max 2,500 tokens)
- **Output:** ICD-9 codes — 7,000 diagnosis + 2,000 procedure (~9,000 total)
- **Embeddings:** 100-dim word2vec pretrained on corpus

	MIMIC-III full
# training documents	47,724
Vocabulary size	51,917
Mean tokens / document	1,485
Mean labels / document	15.9
Total labels	8,922

Baselines

- Single-layer CNN (Kim, 2014)
- Logistic Regression (bag-of-words)
- Bidirectional GRU (BiGRU)

Evaluation Metrics

- Macro / Micro AUC and F1
- Precision@n — fraction of top-n predicted codes that are correct

Experimental Setup - Resume

Each document is a long medical narrative; the model must simultaneously predict ∼16 codes on average, chosen from a space of nearly 9,000 — making this an extremely sparse multi-label problem.

🧉 Results 1/2 ---

Model	AUC Macro	AUC Micro	F1 Macro	F1 Micro	F1 Diag	F1 Proc	P@n 8	P@n 15
Scheurwegs et. al (2017)	–	–	–	–	0.428	0.555	–	–
Logistic Regression	0.561	0.937	0.011	0.272	0.242	0.398	0.542	0.411
CNN	0.806	0.969	0.042	0.419	0.402	0.491	0.581	0.443
Bi-GRU	0.822	0.971	0.038	0.417	0.393	0.514	0.585	0.445
CAML	0.895	**0.986***	**0.088**	**0.539***	**0.524***	**0.609***	**0.709***	**0.561***
DR-CAML	**0.897**	0.985	0.086	0.529	0.515	0.595	0.690	0.548

- CAML gives strongest results on all metrics; DR-CAML is the regularized version ($\lambda = 0.01$)
- Bi-GRU comparable to vanilla CNN; Logistic Regression worse than all neural architectures

🧉 Results - Micro vs. Macro Metrics ---

Model	AUC		F1				P@n	
	Macro	Micro	Macro	Micro	Diag	Proc	8	15
Scheurwegs et. al (2017)	–	–	–	–	0.428	0.555	–	–
Logistic Regression	0.561	0.937	0.011	0.272	0.242	0.398	0.542	0.411
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Bi-GRU	0.822	0.971	0.038	0.417	0.393	0.514	0.585	0.445
CAML	0.895	**0.986***	**0.088**	**0.539***	**0.524***	**0.609***	**0.709***	**0.561***
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- **Macro:** computes the metric separately for each code then averages — every code counts equally, regardless of frequency. A very rare code weighs the same as a very common one.
- **Micro:** pools all true positives, false positives, and false negatives across all codes before computing — frequent codes dominate the result.

In this problem the difference is critical

A model that ignores rare codes can still achieve high Micro-F1, but will be heavily penalized by Macro-F1.

🧉 Interpretability Evaluation ---

- Sample of 100 predicted codes
- Most important k-gram ($k = 4$) + five words on each side for context
- Asked the Physician how informative it is

🧉 Another Baseline: Cosine Similarity Explanation ---

One of the baselines for extracting the text snippet that explains each code.

- Given an ICD code, it computes the cosine similarity between each 4-gram in the document and the code's text description — and returns the most similar one.

It is a purely lexical method: no model involved, just word comparison.

Key Limitation

Works well when the document **literally repeats** words from the code description — **fails** when the relationship is implicit or uses different terminology.

🧉 Interpretability: Qualitative Examples ---

934.1: "Foreign body in main bronchus"

CAML (HI)	*...line placed bronchoscopy performed showing* **large mucus plug on** *the left on transfer to...*
Cosine Sim	*...also needed medication to help* **your body maintain your** *blood pressure after receiving iv...*
CNN	*...found to have a large* **lll lingular pneumonia on** *chest x ray he was...*
Logistic Regression	*...impression confluent consolidation involving nearly* **the entire left lung** *with either bronchocentric or vascular...*

442.84: "Aneurysm of other visceral artery"

CAML (I)	*...and gelfoam embolization of right* **hepatic artery branch pseudoaneurysm** *coil embolization of the gastroduodenal...*
Cosine Sim	*...coil embolization of the gastroduodenal* **artery history of present** *illness the pt is a...*
CNN	*...foley for hemodynamic monitoring and* **serial hematocrits angio was** *performed and his gda was...*
Logistic Regression (I)	*...and gelfoam embolization of right* **hepatic artery branch pseudoaneurysm** *coil embolization of the gastroduodenal...*

428.20: "Systolic heart failure, unspecified"

CAML	*...no mitral valve prolapse moderate* **to severe mitral regurgitation** *is seen the tricuspid valve...*
Cosine Sim	*...is seen the estimated pulmonary* **artery systolic pressure is** *normal there is no pericardial...*
CNN	*...and suggested starting hydralazine imdur* **continue aspirin arg admitted** *at baseline cr appears patient...*
Logistic Regression (HI)	*...anticoagulation monitored on tele pump* **systolic dysfunction with ef** *of seen on recent echo...*

🧉 How to Pick the Most Influential 4-gram? ---

- **CAML**: looks at SoftMax output $\boldsymbol{\alpha}_\ell$ to get the best 4-gram for code ℓ
- **Logistic Regression**: sum of weight matrix coefficients for ℓ over words in k-gram; top-scoring k-gram returned
- **Code descriptions**: cosine similarity between each stemmed k-gram and stemmed ICD-9 code description
- **Max-Pooling CNN**:

$$a_i = \arg \max_{j \in \{1,\ldots,m-k+1\}} (H_{ij}), \qquad \alpha_{i\ell} = \sum_{j:\, a_j = i} \beta_{\ell,j}$$

Select the most important k-gram as $\arg\max_i \alpha_{i\ell}$

🧉 Qualitative Evaluation Results ---

Method	Informative	Highly informative
CAML	46	**22**
Code Descriptions	48	20
Logistic Regression	41	18
CNN	36	13

- CAML selects the greatest number of "highly informative" explanations
- CAML selects more "informative" explanations than both CNN and logistic regression
- Cosine Similarity also performs well; examples in Table 1 demonstrate strengths of CAML

🧉 Conclusion ---

Concluding Thoughts

- Yields strong improvements over previous metrics across several formulations of ICD-9 code prediction
- Provides satisfactory explanations for its predictions per stakeholder evaluation

Future Work

- **Extensible** to other multi-label document tagging tasks
- **Linguistic perspective**: integrate document structure of discharge summaries; better handle out-of-vocabulary tokens
- **Application perspective**: leverage hierarchy of ICD codes; predict codes for future visits

🧉 Discussion (Paper 1) ---

1. What are potential limitations of CAML?
2. The explanations from the attention mechanism are extracted text snippets. Does this provide enough information? What else could be provided?
3. Strengths and weaknesses of their evaluation and results (accuracy and interpretability)
4. What additional applications could CAML be useful for?

🧉 Paper 2 ---

Attention is *not* Explanation

Authors: Sarthak Jain, Byron C. Wallace

Presenters: Nikhil Nayak, Sree Harsha Tanneru, Hongjin Lin

2023/03/06

[@jain2019attention]

🧉 Attention Mechanism: a "Recent" Breakthrough in NLP ---

- **2015**: birth of attention mechanism in NLP
  - "Neural machine translation by jointly learning to align and translate" (Bahdanau et al., 2015)
- **2017**: birth of transformers based on self-attention mechanism
  - "Attention is all you need" (Vaswani et al., 2017)

🧉 Many Use Attention as an Explanation Mechanism ---

🧉 But Does Attention Provide Faithful Explanations? ---

Despite very different attention weights, both yield effectively the same prediction (0.01).

🧉 Research Questions ---

What is the degree to which attention weights provide meaningful "explanations" for predictions?

1. **Consistency**: To what extent do induced attention weights **correlate with measures of feature importance** — specifically, those from gradients and leave-one-out (LOO) methods?
2. **Counterfactual attention weights**: Would **alternative attention weights** (and hence distinct heatmaps/"explanations") necessarily yield different predictions?

🧉 Contributions and Main Findings ---

1. Standard attention modules **do not provide meaningful explanations** and should not be treated as though they do.
2. Learned attention weights are frequently **uncorrelated with gradient-based** measures of feature importance.
3. One can identify very **different attention distributions** that nonetheless yield equivalent predictions.

🧉 Experimental Setup ---

🧉 Correlation Between Attention and Feature Importance ---

To what extent do induced attention weights correlate with measures of feature importance?

Specifically two methods are considered:

1. Feature gradient methods
2. Leave-One-Out (LOO) method

🧉 Gradient-Based Feature Importance ---

- Compute the gradient of the output with respect to each input feature
- This tells us how much changing each feature would affect the output
- Visualize by highlighting input features with highest absolute gradient values

🧉 Feature Erasure / LOO ---

- Remove one input feature at a time and observe how the output changes
- This allows us to measure the impact of each feature on the output
- Visualize by highlighting the removed feature and showing the change in output

🧉 Distribution Change Measure ---

Total Variation Distance (TVD), Kendall's τ coefficient

$$\mathrm{TVD}(\hat{y}_1, \hat{y}_2) = \frac{1}{2} \sum_{i=1}^{|\mathcal{Y}|} |\hat{y}_{1i} - \hat{y}_{2i}|$$

$$\tau = \frac{(\text{concordant pairs}) - (\text{discordant pairs})}{(\text{number of pairs})} = 1 - \frac{2\,(\text{discordant pairs})}{\binom{n}{2}}$$

🧉 Algorithm for Feature Importance Computations ---

$$\mathbf{h} \leftarrow \mathrm{Enc}(\mathbf{x}), \quad \hat{\alpha} \leftarrow \mathrm{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$$

$$\hat{y} \leftarrow \mathrm{Dec}(\mathbf{h}, \alpha)$$

$$g_t \leftarrow \left| \sum_{w=1}^{|V|} \mathbf{1}[\mathbf{x}_{tw} = 1] \frac{\partial y}{\partial \mathbf{x}_{tw}} \right|, \quad \forall t \in [1, T]$$

$$\tau_g \leftarrow \text{Kendall-}\tau(\alpha, g)$$

$$\Delta\hat{y}_t \leftarrow \mathrm{TVD}(\hat{y}(\mathbf{x}_{-t}), \hat{y}(\mathbf{x})), \quad \forall t \in [1, T]$$

$$\tau_{\mathrm{loo}} \leftarrow \text{Kendall-}\tau(\alpha, \Delta\hat{y})$$

🧉 Feature Gradients Results ---

Histogram of Kendall correlation between attention and gradients.

🧉 Correlation Comparison: Attention, Gradients, LOO ---

🧉 Possible Limitations in Experiment Results ---

1. Potential issues with Kendall τ correlation — many irrelevant features may add noise to the correlation measures
2. Alternative measures of feature importance (gradients and LOO) may not be considered ideal or the 'ground truth' for comparison purposes

Counterfactual Attention Weights ---

Would the prediction be different if the model attended to different input features?

1. **Consistency**: correlation with other measures of feature importance
2. **Counterfactual attention distributions** should yield corresponding changes in predictions

🧉 Counterfactual Attention: Two Experiments ---

1. **Attention Permutation**: scramble the original attention weights
2. **Adversarial Attention**: generate an adversarial distribution that is maximally distinct from the observed attention weights, whilst simultaneously close to the observed prediction

🧉 Attention Permutation: Algorithm ---

$\mathbf{h} \leftarrow \mathrm{Enc}(\mathbf{x}),\ \hat{\alpha} \leftarrow \mathrm{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$
$\hat{y} \leftarrow \mathrm{Dec}(\mathbf{h}, \hat{\alpha})$
for $p \leftarrow 1$ to 100 **do**
    $\alpha^p \leftarrow \mathrm{Permute}(\hat{\alpha})$
    $\hat{y}^p \leftarrow \mathrm{Dec}(\mathbf{h}, \alpha^p)$ ▷ $\mathbf{h}$ not changed
    $\Delta\hat{y}^p \leftarrow \mathrm{TVD}[\hat{y}^p, \hat{y}]$
end for
$\Delta\hat{y}^{med} \leftarrow \mathrm{Median}_p(\Delta\hat{y}^p)$

- **h** (the hidden state) is kept fixed
- Only the attention weights are permuted
- Measures how much the output changes when attention is scrambled

Attention Permutation: Results ---

Adversarial Attention ---

- Explicitly seek attention weights that differ as much as possible from observed weights, yet leaving prediction almost unchanged
- **Why?** Alternative attention distributions for the same output may be viewed as equally plausible explanations:
  - Cannot conclude the model made a specific prediction because it attended in a certain way
  - Lack of specificity makes constructing counterfactual explanations hard
  - Makes model less trustworthy

🧉 Adversarial Attention: Optimisation ---

$$JSD(\alpha_1, \alpha_2) = \frac{1}{2}\left(KL\left[\alpha_1 \,\middle|\, \frac{\alpha_1 + \alpha_2}{2}\right] + KL\left[\alpha_2 \,\middle|\, \frac{\alpha_1 + \alpha_2}{2}\right]\right)$$

Seek $\alpha^{(1)}, \ldots, \alpha^{(k)}$ that maximize f subject to small output change:

$$\underset{\alpha^{(1)},\ldots,\alpha^{(k)}}{\text{maximize}}\ f\left(\{\alpha^{(i)}\}_{i=1}^{k}\right) \quad \text{s.t.} \quad \forall i\ \text{TVD}[\hat{y}(\mathbf{x}, \alpha^{(i)}), \hat{y}(\mathbf{x}, \hat{\alpha})] \leq \epsilon$$

$$f\left(\{\alpha^{(i)}\}_{i=1}^{k}\right) = \sum_{i=1}^{k} JSD[\alpha^{(i)}, \hat{\alpha}] + \frac{1}{k(k-1)} \sum_{i<j} JSD[\alpha^{(i)}, \alpha^{(j)}]$$

🧉 Adversarial Attention: Algorithm ---

$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}),\ \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$

$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$

$\alpha^{(1)}, \ldots, \alpha^{(k)} \leftarrow \text{Optimize Eq 1}$

for $i \leftarrow 1$ to k **do**

$\quad \hat{y}^{(i)} \leftarrow \text{Dec}(\mathbf{h}, \alpha^{(i)})$ ▷ **h** is not changed

$\quad \Delta\hat{y}^{(i)} \leftarrow \text{TVD}[\hat{y}, \hat{y}^{(i)}]$

$\quad \Delta\alpha^{(i)} \leftarrow JSD[\hat{\alpha}, \alpha^{(i)}]$

end for

$\epsilon\text{-max JSD} \leftarrow \max_i \mathbf{1}[\Delta\hat{y}^{(i)} \leq \epsilon]\, \Delta\alpha^{(i)}$

🧉 Adversarial Attention: Results ---

🧉 Adversarial Attention: Results ---

Counterfactual Attention: Limitations ---

- Adversarial weights may themselves be unlikely under model parameters
- There are instances where a single explanation out of multiple plausible explanations is sufficient

🧉 Overall Limitations ---

- Alternate attention mechanisms are not explored
- Limited evaluations to classification, QA, and NPI tasks; seq2seq tasks are not considered

🧉 Critique and Discussion ---

- Wiegreffe & Pinter (2019), *Attention is not not Explanation*, EMNLP-IJCNLP:
  - Attention distribution is not a primitive
  - Is attention necessary for prediction?
  - Is the variance in attention distributions unusual? i.e., How adversarial are the adversaries?
- Do you agree with the authors' conditions of a "faithful explanation"?
- Do you think attention constitutes a "sufficient" explanation?
- What other conditions should we consider as a good explanation?
- How can experts be brought in to evaluate conditions of good explanations?

Thank You!

🧉 References --- |

Mullenbach, James, Sarah Wiegreffe, Jon Duke, Jimeng Sun, and Jacob Eisenstein. 2018. "Explainable Prediction of Medical Codes from Clinical Text." *arXiv Preprint arXiv:1802.05695*.